Python Activity-Based Learning
Program 9: Multi-Currency Travel Budget Analysis

Problem Statement

Convert travel expenses from different currencies into Indian rupees using supplied exchange rates and conversion fees. Compare total spending with a budget and identify the largest expense category.

Objectives

• Convert each expense to Indian rupees.

• Add a currency conversion fee.

• Compare total spending with the travel budget and identify the largest expense.

Assumptions

• The user enters the exchange rate as INR for one unit of foreign currency.

• Conversion fee is 2% of the converted amount.

• All expenses are positive or zero; budget is in INR.

Algorithm

1. Read travel budget and number of expense categories.

2. For each category, read its name, amount in foreign currency, and exchange rate.

3. Convert the amount to INR and add a 2% fee.

4. Add each converted expense to the total and track the largest category.

5. Compare total spending with budget.

6. Display category-wise costs, total, and budget status.

Flowchart

```
START → Read budget and categories → Read expense and exchange rate → Convert to INR → Add 2% fee → Update tot
```

Pseudocode

```
BEGIN
  READ budget, n
  total = 0
  largest = 0
  FOR each category
    READ name, amount, rate
    converted = amount * rate
    fee = converted * 2 / 100
    cost = converted + fee
    total = total + cost
    IF cost > largest THEN save category as largest
  END FOR
  DISPLAY total and largest category
  IF total <= budget THEN DISPLAY within budget
  ELSE DISPLAY over budget
END
```

Python Source Code

```
budget = float(input("Enter travel budget in INR: "))
n = int(input("Enter number of expense categories: "))

total = 0
largest_cost = -1
largest_category = ""

for i in range(n):
    print("\nExpense", i + 1)
```

```
    category = input("Category name: ")
    amount = float(input("Amount in foreign currency: "))
    rate = float(input("Exchange rate (INR per 1 unit): "))

    if amount < 0 or rate <= 0:
        print("Invalid amount or exchange rate.")
        continue

    converted = amount * rate
    fee = converted * 0.02
    cost = converted + fee
    total = total + cost

    print("Cost including fee: Rs.", round(cost, 2))

    if cost > largest_cost:
        largest_cost = cost
        largest_category = category

print("\nTRAVEL BUDGET REPORT")
print("Total expenses: Rs.", round(total, 2))
print("Largest expense category:", largest_category)

if total <= budget:
    print("You are within the travel budget.")
else:
    print("You have exceeded the travel budget by Rs.",
          round(total - budget, 2))
```

Test Case

Budget ■50,000. Hotel: 200 USD at ■83/USD. Food: 100 EUR at ■90/EUR.
Expected: Hotel ■16,932; Food ■9,180; total ■26,112; largest category Hotel; within budget.

Sample Output

```
Cost including fee: Rs. 16932.0
Cost including fee: Rs. 9180.0
TRAVEL BUDGET REPORT
Total expenses: Rs. 26112.0
Largest expense category: Hotel
You are within the travel budget.
```

Conclusion

The program converts foreign-currency expenses into INR, includes a conversion fee, identifies the largest expense category, and checks the travel budget.